

Duality

Properties and Applications

Marc Sunet

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Table of contents

Table of Contents

Motivation

Why do we need duality?

The applet

<http://www-ma2.upc.es/~geoc/dual/dual.html>

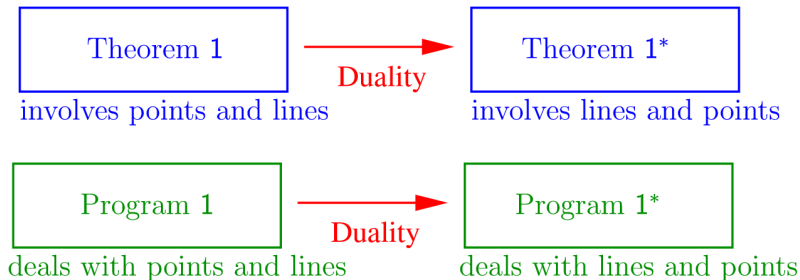
Duality in the plane

A function $D : \mathbb{R}^2 \rightarrow \mathbb{R}^2$ that transforms points to lines and lines to points.

Several maps exist, each with their own properties and characteristics.

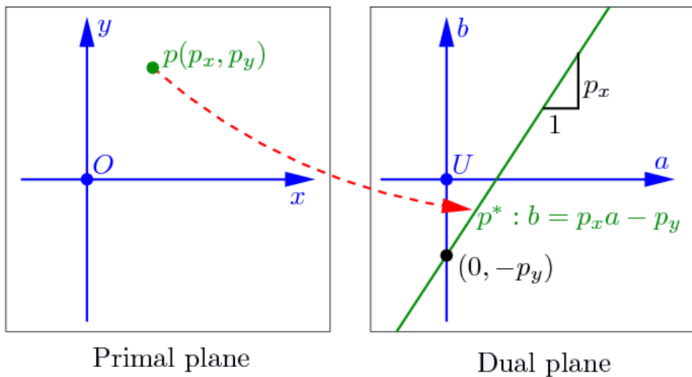
Different maps to solve different problems.

Duality of theorems



Standard duality

$$p = (a, b) \mapsto p^* : y = ax - b$$
$$l : y = mx + n \mapsto l^* = (m, -n)$$



Standard duality

$$\begin{aligned} p = (a, b) &\mapsto p^* : y = ax - b \\ l : y = mx + n &\mapsto l^* = (m, -n) \end{aligned}$$

Vocabulary:

Primal space The original space.

Dual space The original space transformed by D .

Essential properties:

1. D is a bijection between points and non-vertical lines.
 $\rightarrow$ "we can go back".
2. $D^2 = Id$
 $\rightarrow$ "to go back we just apply D again".

Parallel lines

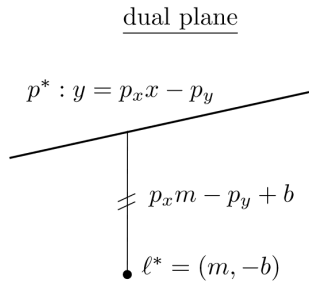
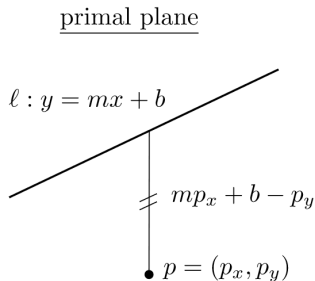
Points sharing the same x coordinate map to parallel lines:

$$(a, b) \mapsto y = ax - b$$

$$(a, c) \mapsto y = ax - c$$

Distance preservation

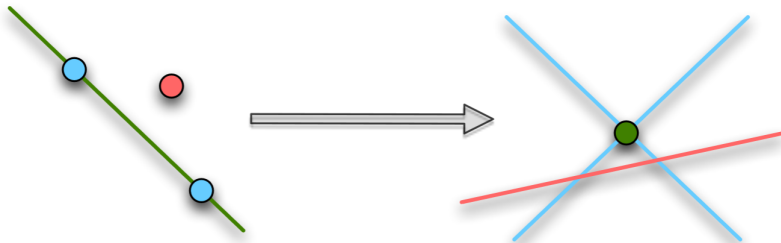
Duality preserves point-line vertical distances:



$\Rightarrow$ Preserves incidence: $p \in \ell \Leftrightarrow \ell^* \in p^*$

$\Rightarrow$ Preserves order: p above $\ell \Leftrightarrow \ell^*$ above p^*

Order preservation



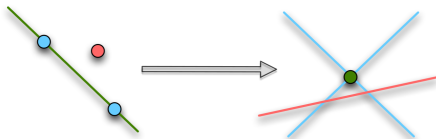
Red above green $\Leftrightarrow$ green* above red*

Order preserving or reversing?

Both!

Order preserving:

- ▶ p above $l \Leftrightarrow l^*$ above p^*

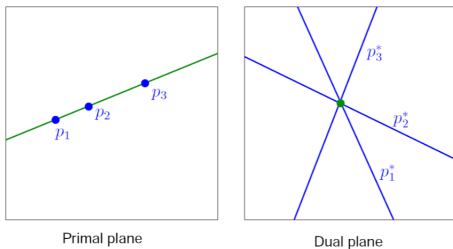


Order reversing:

- ▶ p above $l \Leftrightarrow p^*$ below l^*

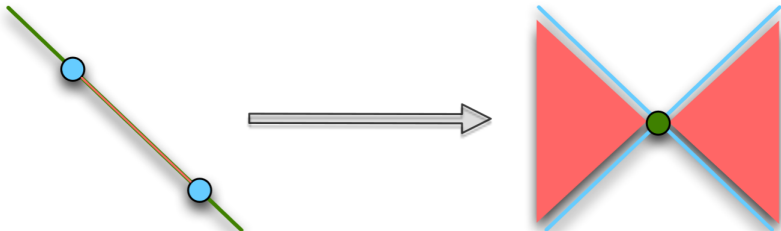
Collinearity

Collinear points dualise into lines sharing a common intersection point (and vice versa):



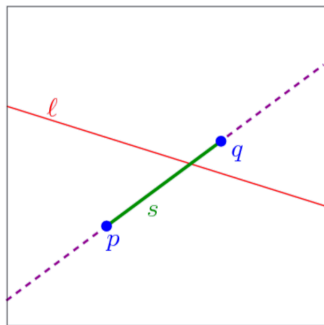
$$\left. \begin{array}{l} p_1 \in l \Rightarrow l^* \in p_1^* \\ p_2 \in l \Rightarrow l^* \in p_2^* \\ \dots \\ p_N \in l \Rightarrow l^* \in p_N^* \end{array} \right\} \Rightarrow l^* \text{ is the common intersection point of all } N \text{ lines}$$

Duality of a line segment

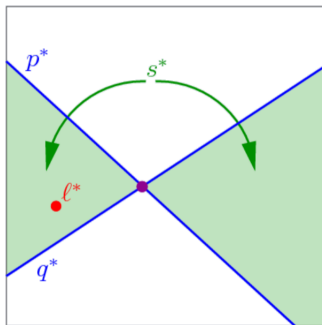


- ▶ Endpoints map to two intersecting lines.
- ▶ Intermediate points are all collinear $\Rightarrow$ share common intersection point.
- ▶ The dual of intermediate points are lines that sweep from one dual endpoint to the other.

Line-Segment intersection



Primal plane

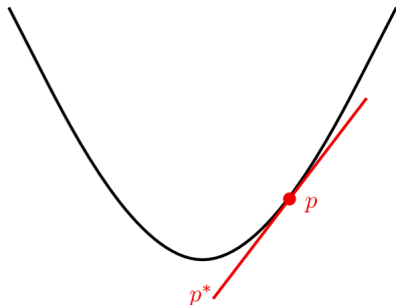


Dual plane

- ▶ Line l intersecting s has p or q above it and the other point below it.
- ▶ Order preservation $\Rightarrow p^*$ or q^* has the point l^* above it and the other line below it.
- ▶ $\Rightarrow l^*$ lies inside the double-wedge s^* .

Duality and the $\frac{x^2}{2}$ parabola

Consider parabola $P : y = \frac{x^2}{2}$:



$$p \in P \Rightarrow p^* \text{ tangent to } P \text{ at } p$$

Duality and the $\frac{x^2}{2}$ parabola

p^* parallel to tangent at p :

- ▶ Tangent $y'(x) = x$, $y'(a) = a$
- ▶ $p^* \equiv y = ax - \frac{a^2}{2}$
- ▶ $y'(x)$ and p^* have the same slope

p^* contains p :

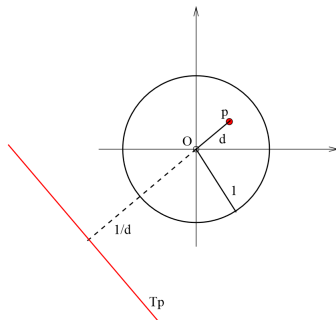
- ▶ $p = (a, b) \in P \Rightarrow p = (a, \frac{a^2}{2})$
- ▶ $p^* \equiv y = ax - \frac{a^2}{2}$
- ▶ $\frac{a^2}{2} = a^2 - \frac{a^2}{2}$

$\Rightarrow p^*$ tangent to P at p

Polar Duality

$$(a, b) \mapsto ax + by + 1 = 0$$

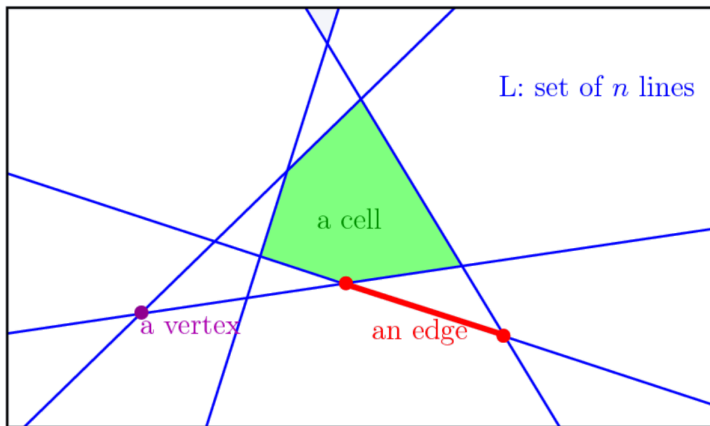
$$ax + by + 1 = 0 \mapsto (a, b)$$



If d is the distance from the origin O to p , then p^* is the line perpendicular to Op at distance $1/d$ from O and placed on the other side of O .

Table of Contents

Arrangements



arrangement $\mathcal{A}(L)$

Simple arrangements

Simple arrangement:

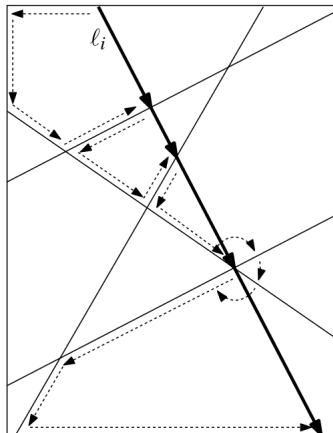
- ▶ No two lines are parallel.
- ▶ No three lines pass through the same point.

Combinatorial complexity:

- ▶ $\# \text{Vertices} \leq n(n-1)/2$
- ▶ $\# \text{Edges} \leq n^2$
- ▶ $\# \text{Faces} \leq n^2/2 + n/2 + 1$

Overall complexity: $\mathcal{O}(n^2)$

Constructing arrangements



Algorithm CONSTRUCTARRANGEMENT(L)

Input. Set L of n lines

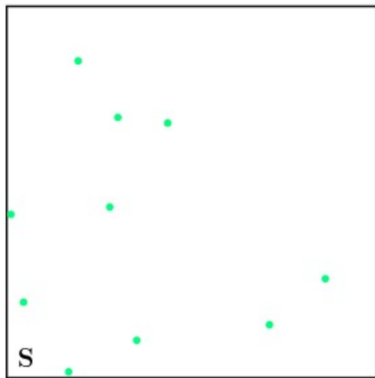
Output. DCEL for $\mathcal{A}(L)$ in $\mathcal{B}(L)$

1. Compute bounding box $\mathcal{B}(L)$
2. Construct DCEL for subdivision induced by $\mathcal{B}(L)$
3. **for** $i \leftarrow 1$ **to** n
4. **do** insert ℓ_i

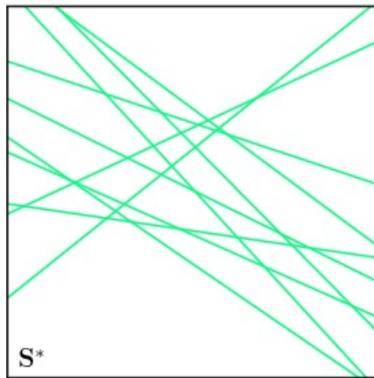
Cost: $\mathcal{O}(n^2)$

Arrangements and duality

Why are arrangements important?



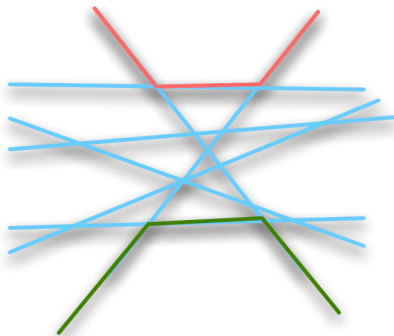
Primary plane



Dual plane

Table of Contents

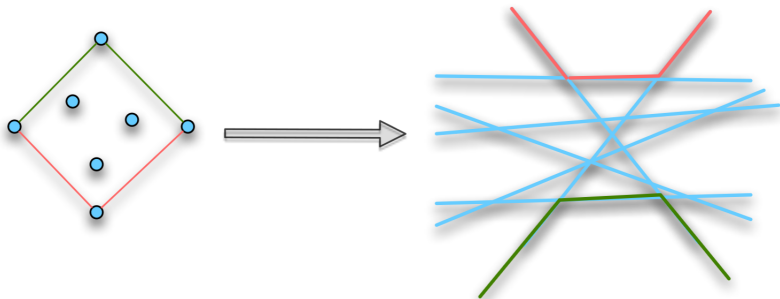
Envelopes



Upper envelope of a set of lines is the set of points that are above all lines.

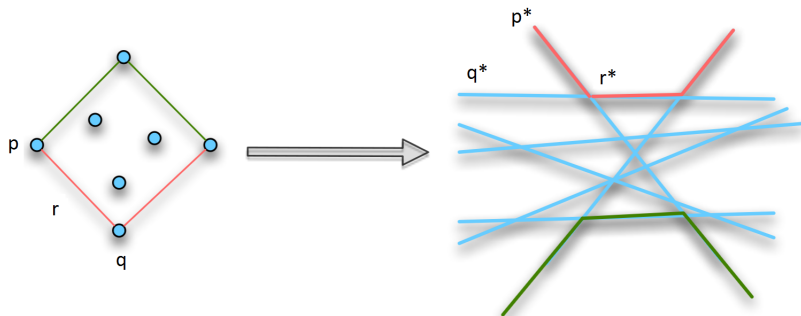
Lower envelope of a set of lines is the set of points that are below all lines.

Convex Hulls



- ▶ We can compute upper and lower envelopes by dualising a convex hull.
- ▶ Or we can compute the convex hull by dualising the upper and lower envelopes.

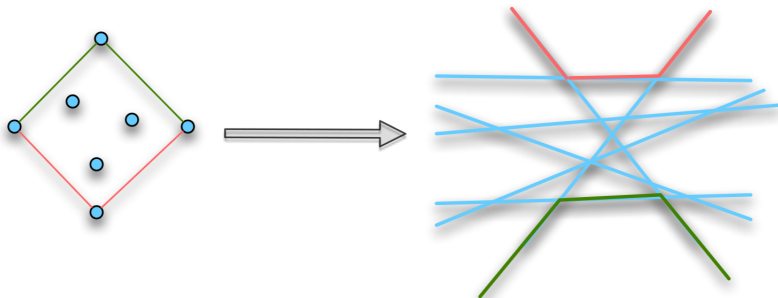
Lower hull / upper envelope



Consider lower hull of points (red):

- ▶ Supporting lines l_i of lower hull lie on or below all points p_j .
- ▶ Order preserving $\Rightarrow$ dual points l_i^* of upper envelope lie on or above all dual lines p_j^* .
- ▶ Incidence $\Rightarrow l_i^*$ are the intersection points of the dual lines p_j^* of the upper envelope.

Upper hull / lower envelope



Consider upper hull of points (green):

- ▶ Supporting lines l_i of upper hull lie on or above all points p_j .
- ▶ Order preserving $\Rightarrow$ dual points l_i^* of lower envelope lie on or below all dual lines p_j^* .
- ▶ Incidence $\Rightarrow l_i^*$ are the intersection points of the dual lines p_j^* of the lower envelope.

Convex Hull - Envelope Duality

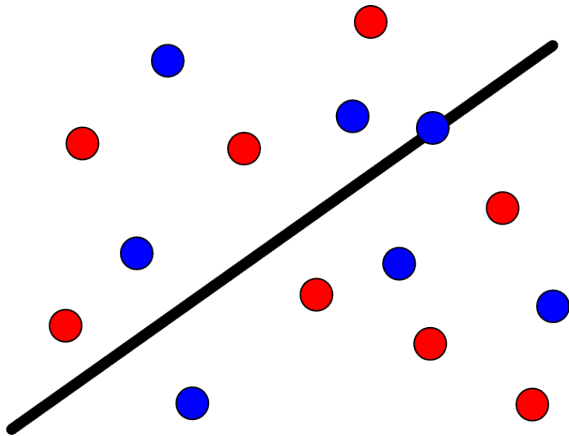
Constructing the convex hull $CH(P)$ of a set P of n points takes $\mathcal{O}(n \log n)$ time.

Checking whether a point $p \in P$ belongs to $CH(P)$ takes $\mathcal{O}(n)$ time:

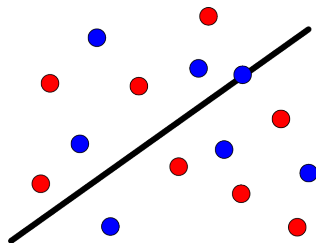
$$\blacktriangleright p \in CH(P) \Leftrightarrow p^* \in \text{Envelopes}(D(P))$$

Table of Contents

Ham Sandwich Cuts



Ham Sandwich Cuts



Ham sandwich theorem (measure theory):

- ▶ Given n measurable "objects" in n -dimensional space, it is possible to divide all of them in half (wrt their measure, i.e. volume) with a single $(n-1)$ -dimensional hyperplane.
- ▶ Object: set of finite measure, so that the notion of "dividing its volume in half" makes sense.

Ham Sandwich Cuts



Ham Sandwich Cuts

The theorem guarantees that such a cut exists.

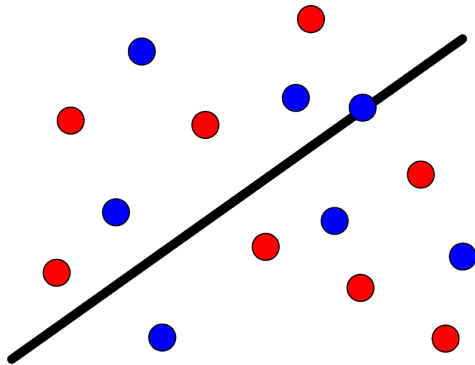
- ▶ But finding it is generally an unsolved problem.

Fortunately, well-known algorithms exist for $2d$ and $3d$.

Also note:

- ▶ The ham sandwich cut *bisects* the n input objects.
- ▶ Finding the line/curve that completely separates them is another problem known as the Separability Problem.

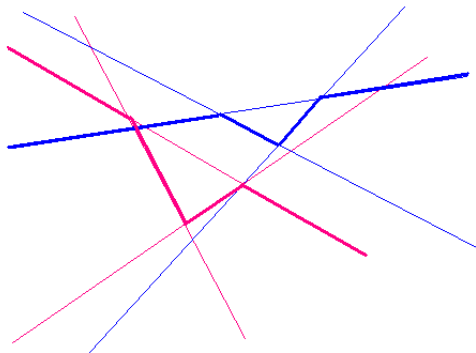
Ham Sandwich Cuts: $2D$ discrete case



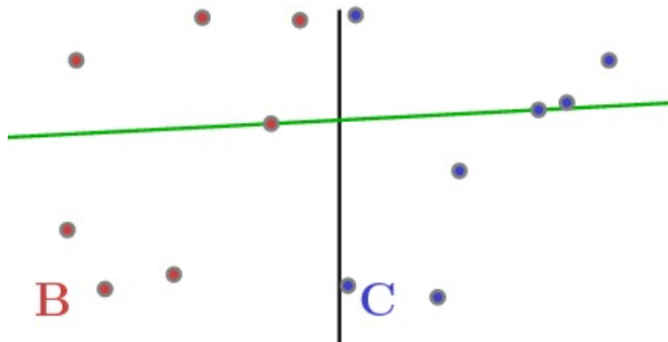
- ▶ Given two sets of points.
- ▶ Find line that simultaneously splits both sets in equal halves.

Ham Sandwich Cuts: Median Level

The **median level** of a set of lines is a "line" (set of line segments) that has half the lines above and half the lines below.



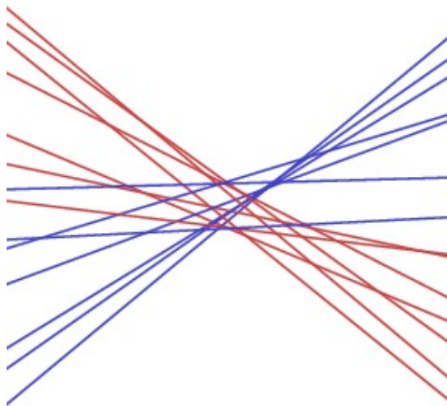
Ham Sandwich Cuts: Simple case



- ▶ Given two linearly separable sets of points B, C .
- ▶ Find line that splits both B and C in two equal halves.

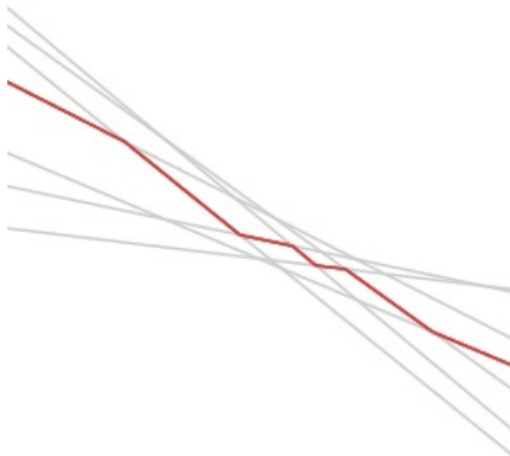
Ham Sandwich Cuts: Simple case

Step 1: Dualise B, C to obtain B^*, C^* :



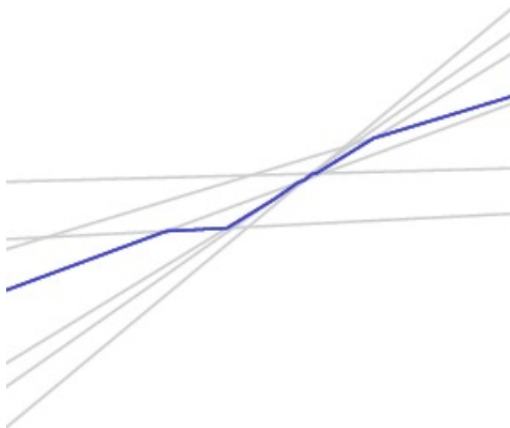
Ham Sandwich Cuts: Simple case

Step 2: Compute the median level of B^* :



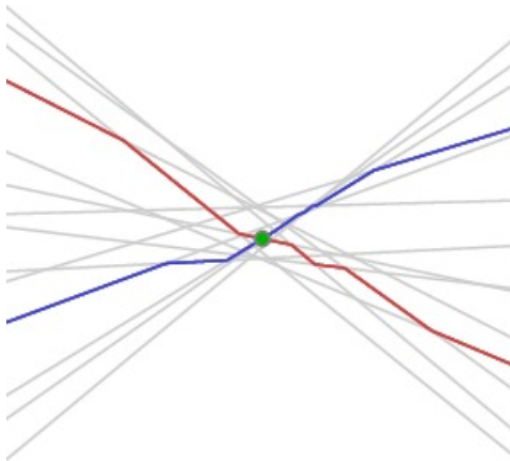
Ham Sandwich Cuts: Simple case

Step 3: Compute the median level of C^* :



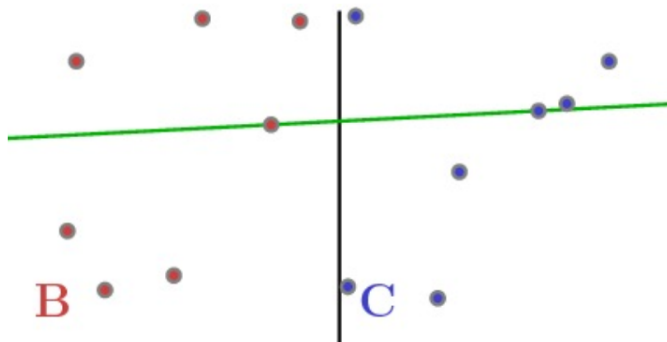
Ham Sandwich Cuts: Simple case

Step 4: Intersect the two median levels:



Ham Sandwich Cuts: Simple case

Step 5: Dualise the intersection points. This yields the ham sandwich cut:



Ham Sandwich Cuts: General case

What about the general case?

- ▶ Prune-and-search algorithm that solves the problem in $\mathcal{O}(n)$ time.
- ▶ At each iteration, a constant number of lines that do not have intersections in the median level is discarded.
- ▶ The algorithm repeats this process until a small number of lines remain, in which case the problem can be solved directly.

Table of Contents

Duality in 3D

In 3D we have point-plane duality.

- ▶ Points map to planes.
- ▶ Planes map to points.
- ▶ Lines map to lines.

An arrangement of n planes has complexity $\mathcal{O}(n^3)$.

Duality in Higher Dimensions

Points map to hyperplanes, hyperplanes map to points:

$$\begin{aligned} p = (p_1, p_2, \dots, p_d) &\mapsto x_d = p_1x_1 + p_2x_2 + \dots + p_{d-1}x_{d-1} - p_d \\ x_d = a_1x_1 + a_2x_2 + \dots + a_{d-1}x_{d-1} + a_d &\mapsto (a_1, a_2, \dots, -a_d) \end{aligned}$$

An arrangement of n hyperplanes in d -dimensional space has complexity $\mathcal{O}(n^d)$.

Table of Contents

Links and References

Computational Geometry: Algorithms and Applications (Mark de Berg, Otfried Cheong, Marc van Kreveld, Mark Overmars)

Intersecting Half-Planes and Related Problems (Vera Sacristán)

<http://www-ma2.upc.es/vera/wp-content/uploads/2013/06/IntersectingHalfplanes.pdf>

Arrangements and Duality (Partha P. Goswami)

<http://www.cs.uu.nl/docs/vakken/ga/slides8.pdf>

Duality Transformation and its Applications to Computational Geometry (Partha P. Goswami)

<http://www.tcs.tifr.res.in/~ghosh/partha1-lecture.pdf>

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<http://algo.kaust.edu.sa/Documents/cs372l13.pdf>

Links and References

Geoduality and Linear Programming (R Sharathkumar and Harish Chandran)

<https://www.cs.duke.edu/~harish/papers/geoduality.pdf>

Arrangements and Duality (Suresh Venkatasubramanian)

<http://www.cs.utah.edu/~suresh/compgeom/arrangements.pdf>

Finding $2d$ Ham Sandwich Cuts in Linear Time (Benjamin Armbruster)

<http://users.iems.northwestern.edu/~armbruster/hamsandwich.pdf>

Creating a Ham Sandwich Cut (Danielle MacNevin)

<http://cgm.cs.mcgill.ca/~athens/cs507/Projects/2002/DanielleMacNevin/algorithm-pg1.html>

Ham Sandwich Cuts (Rudolf Fleischer)

http://www.tcs.fudan.edu.cn/rudolf/Courses/Algorithms/Alg_cs_07w/Webprojects/Jiajin_redblue/intro.html